

세라믹개론(공) (MSE309-01)

학기	2026년 2학기
담당교수	이태훈
이수구분	교선평수
학점	3.0
강의시간	월10:30-12:00(E7 - 307), 수10:30-12:00(E7 - 307)
강의실	E6 414 (교수실)
소속	기초학부

교과개요

This introductory course provides fundamental knowledge for understanding ceramic materials, with emphasis on advanced ceramics. Beginning with the definition and classification of ceramics, the course examines atomic bonding, crystal structures, defect chemistry, diffusion, and phase stability. Students will learn how these physicochemical phenomena govern the dielectric, magnetic, optical, and related functional properties of ceramics, and how structure–property relationships can be interpreted from a materials science perspective. In addition, the course includes computer-assisted visualization of materials structures, computer-based thermodynamic analysis and phase-diagram plotting, and visualization and analysis of ceramic-property data.

교과목표

After completing this course, students will be able to: (1) explain the definition, classification, and major categories of ceramics; (2) understand fundamental physicochemical phenomena in ceramics, including defects, nonstoichiometry, and atomic transport; (3) relate crystal structure, defect chemistry, diffusion, and phase stability to ceramic properties; (4) explain dielectric, magnetic, and optical properties from the viewpoint of ceramic structure and bonding; (5) use computer tools to visualize ceramic structures and interpret thermodynamic trends, phase diagrams, and property data; and (6) analyze advanced ceramics using a structure–property framework grounded in the fundamentals of ceramic materials science.

주차별 강의계획

- 1주차: Introduction to Ceramics: Definition, classification, and scope of ceramics; overview of advanced ceramics
- 2주차: Basics on Ceramics: General characteristics of ceramics and introductory concepts in ceramic materials
- 3주차: Physics and Structure I: Review of basic materials science and structural features of ceramics
- 4주차: Physics and Structure II: Crystal structure, bonding, and fundamental physicochemical properties (with tutorial session)
- 5주차: Physics and Structure III: Continuation of structural features and basic physicochemical behavior of ceramics
- 6주차: Defect Chemistry I: Point defects, nonstoichiometry, and defect formation in ceramics (with tutorial session)
- 7주차: Defect Chemistry II: Charge compensation and the relationship between defects and properties
- 8주차: Midterm Examination
- 9주차: Diffusion I: Introduction to diffusion and atomic/ionic transport in ceramics (with tutorial session)
- 10주차: Diffusion II: Diffusion mechanisms and diffusion coefficients
- 11주차: Diffusion III: Further discussion of diffusion and its implications for ceramic behavior
- 12주차: Phase Stability I: Phase stability and phase transformation in ceramics
- 13주차: Phase Stability II: Continuation of phase stability and related phenomena (with tutorial session)
- 14주차: Dielectric, Magnetic, and Optical Properties I: Structure–property relationships in dielectric, magnetic, and

optical ceramics

15주차: Dielectric, Magnetic, and Optical Properties II: Structure–property relationships in dielectric, magnetic, and optical ceramics

16주차: Final Examination

수업방법

Lecture and tutorial session with your laptop or computer

학습윤리

Students are encouraged to discuss course materials with their peers while studying. However, all submitted assignments must be prepared through each student's own effort.

교과정책(방침)

Absences without penalty are permitted for specific reasons, including but not limited to: documented medical emergencies, family emergencies, and reserve forces training. If an emergency causes you to miss a graded activity (such as a midterm or final exam), you must contact the instructor immediately to provide documentation and arrange for appropriate make-up work.

준비물 및 기타

Textbook: Fundamentals of ceramics (by M. W. Barsoum) or Korean version